

Aleksandr Petiushko, PhD

Industry and Academia



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Bio

Dr. Aleksandr Petiushko is a Head of AI at [Logical Intelligence](#), a Vice President, Head of RnD at a FinTech Holding, an Adjunct Professor at [Sofia University](#) (Palo Alto, California) giving courses on ML, AI, and the Theory of Deep Learning. Before FinTech, worked as Sr. Director, Head of AI Research at Gatik, Director, Head of ML Research at Nuro, as a Team Lead / Scientific Expert, Chief Scientist at Huawei, as a Managing Director / Leading Scientific Researcher at Artificial Intelligence Research Institute. The Ph.D. dissertation is at the intersection of Discrete Mathematics and Computer Linguistics. Research interests lie in the applications of empirical and theoretical robustness (publications at ECCV, IJCAI, AAAI, CVPR, NeurIPS).

Education

- PhD in Theoretical CS · 2016
- MSc in Mathematics · 2006

Interests

- Financial Technology
- Autonomous Driving
- Deep Learning
- Robustness Theory
- Discrete Mathematics

Skills

- Research
- Leadership
- Lecturing

Summary || Last Update: May 2026

Principal RnD Researcher (20 years of experience), RnD Technical Leader (15 years of experience), and RnD Manager (10 years of experience). Running and managing industrial research and academic collaboration (45 publications, 40 patents). Hiring and transforming AI/ML teams. Inspired by theoretical computer science and how it changes the world.

Experience

Present

Head of AI

Logical Intelligence, USA/CA, San Francisco

Responsibilities:

- Hiring the AI and Research teams
- Roadmapping the AI side of the Company

Technical Directions:

- Latent non-Autoregressive Reasoning
- Non-textual Energy-Based Modeling
- Textual Energy-Based Modeling

Jan 2026

Present

Vice President, Head of RnD FinTech Holding, USA/CA, San Jose

Responsibilities:

- Hiring the Research and Development team
- Research Strategy for the whole FinTech Holding
- Updating infrastructure for the State-of-the-Art ML Research

Technical Directions:

- Predictive tasks (Credit Scoring and Risks, Churn Prediction)
- Recommendation tasks (Personal Finance, Lead Generation)
- Anomaly Detection tasks (Fraud Detection)

Oct 2025

Jun 2025

Head of AI Research

Elea Health, USA/CA, Palo Alto

Responsibilities:

- State-of-the-Art AI Approach to Medical Domain
- Scaling ML infrastructure and environment

Technical Direction: Analysis of ultra-dimensional (billions of pixels) pathology images from microscopy scans

Apr 2025

Apr 2025

Sr. Director, Head of AI Research Gatik AI, USA/CA, Mountain View

Responsibilities:

- Hiring the AI Research team
- AI Strategy
- Scaling ML infrastructure and environment

Technical Directions:

- Online Mapping
- Prediction and Planning
- End-to-end Autonomous Driving Systems
- Uncertainty and Robustness

Sep 2024

Sep 2024

Director, Head of ML Research

NURO, USA/CA, Mountain View

Responsibilities:

- Managing and hiring the team of highly skilled diverse talents in ML (10+ FTEs, dotted line reports, consultants, interns)
- Internal cross-collaboration and ideas brainstorming with other Behavior and Perception teams
- Technical Roadmaps
- State-of-the-Art frontier research
- Academia collaboration

Technical Directions:

- Prediction (including conditional and joint), Planning (including Diffusion-based), and Motion Selection (RL-based)
- Closed-loop Reasoning
- Synthetic Data Generation
- Reward and Issue Predictor models
- Agent-centric and scene-centric encoders
- Scaling laws in Behavior
- Robustness and Uncertainty of Autonomy Stack
- LLM, VLM, Perception-Behavior interface

Feb 2022

Sep 2024

Feb 2022

Feb 2022

Sep 2021

Sep 2021

Sep 2014

May 2014

Nov 2008

Nov 2008

Dec 2003

Director, Head of ML Research

Nuro, USA/CA, Mountain View

Other achievements:

- Internal ML education through a series of AI Theory lectures
- Main ideologist and program owner of Nuro ML University (100+ FTEs involved)
- Created Nuro Tech Talks series
- Made SotA exploration a working internal pipeline
- Constantly increasing Nuro's visibility (participation at conferences, workshops, seminars, etc)

Director, Leading Scientific Researcher

AIRI, Russia, Moscow

Fusion Brain Research Director.

Responsibilities:

- Research roadmap formulation
- Research team hiring
- Leading the research team (6 people)
- Academia collaboration

Technical Directions:

- Multi-modality
- Multi-tasking
- Retrieval-based systems

Chief Scientist, Technical Leader, Research Manager

Huawei Research, Russia, Moscow

Responsibilities:

- Video Intelligence direct management (~15 people, plus interns)
- Fundamental Research (~25 experts from different groups) team leadership
- Academia collaboration projects on Fundamental Research in AI
- Interviewing young talents
- Mentoring interns
- Remote work coordination

Technical Directions:

- Video Recognition and Retrieval (MML, MDMMT, top place on smth2smth)
- Trajectory Prediction (top places on Argoverse and Apolloscape benchmarks)
- Face Recognition (Detection and Identification)
- Domain Adaptation (computer vision)
- Multi-task end2end (keypoints, detection, segmentation)
- Storage (cold storage, erasure coding, object distribution, file type recognition with ML)
- NeuralNets for Computer Vision (Super Resolution, DeBlur, Compression)
- Robustness in DL
- Geometrical DL

Other achievements:

- Organized and maintained the Fundamental Research team (with publishing activity)
- Organized and initially led the first Neural Net direction at Huawei MRC
- Proposed and prepared the multi-year funded project "Fundamental Research in AI"

Senior Software Engineer

LSI Corporation (now Broadcom), USA/CA, San Jose (remote)

Engineering responsibilities:

- Media gateway (DSP for audio)
- WCDMA protocol (preamble detection)
- LDPC (trapping set analysis)
- Gesture recognition (classical ML in CV)
- Flash controller (wear leveler, garbage collector)

Other achievements:

- 30+ patents on storage systems, ML and Computer Vision
- Some demo videos: [1](#), [2](#), [3](#)

Software Engineer

Different Software Companies Russia

Initial steps towards Research and SWE Career.

Education

Mar 2016	Doctor of Philosophy (Ph.D.)	Lomonosov Moscow State University, Russia, Moscow Russian name - "Candidate of Physical and Mathematical Sciences", defended at Lomonosov MSU, Department of Mechanics and Mathematics. Dissertation " Bigram Languages ", major 01.01.09 - Discrete Mathematics and Mathematical Cybernetics. Abstract . Youtube recording (all links are in Russian).
Nov 2012	Postgraduate Student	Lomonosov Moscow State University, Russia, Moscow Postgraduate study at Lomonosov MSU, Department of Mechanics and Mathematics. Major 01.01.09 - Discrete Mathematics and Mathematical Cybernetics.
Sep 2009		
Jul 2006	Master of Science (M.Sc.)	Lomonosov Moscow State University, Russia, Moscow Russian name - "specialist" (theoretical and applied mathematics), major in the discrete mathematics at Lomonosov MSU, Department of Mechanics and Mathematics. Thesis "Dynamic adjustment of signals". Incomplete (but the best that I could find) text (in Russian). GPA 4.89/5, Diploma with honors.
Sep 2000		
Jul 2000	High School	Briansk Pushkin's Lycee, Russia, Briansk High School, Briansk Pushkin's Lycee, Physics and Mathematics major. Gold medal.
Jan 1997		

Academia

Jul 2023 - ...	Adjunct Professor	Sofia University, USA/CA, Palo Alto Machine Learning and Artificial Intelligence courses.
Feb 2023 - ...	Visiting Professor	Moscow Institute of Physics and Technology, Russia, Moscow Theory of Deep Learning courses.
Feb 2019 - ...	Visiting Professor	Lomonosov Moscow State University, Russia, Moscow Python, Computer Vision, Machine Learning, Theory of Deep Learning courses.

Publications

Selected conference/journal/patent publications. For the full list (35+ publications, 30+ patents, about 70 in total), please refer either to the [Google Scholar](#) or [Personal Webpage](#).

2025	J. Zhang, X. Yang, T. Wang, Y. Yao, A. Petiushko, and B. Li. "SafeAuto: Knowledge-Enhanced Safe Autonomous Driving with Multimodal Foundation Models". In: <i>International Conference on Machine Learning (ICML)</i> . 2025
2025	C. Xu, A. Petiushko, D. Zhao, and B. Li. "Diffscene: Diffusion-based safety-critical scenario generation for autonomous vehicles". In: <i>The 39th Annual AAAI Conference on Artificial Intelligence</i> . 2025
2024	J. Booher, K. Rohanimanesh, J. Xu, and A. Petiushko. <i>CIMRL: Combining Imitation and Reinforcement Learning for Safe Autonomous Driving</i> . 2024. arXiv Preprint: 2406.08878
2023	Z. Zhou, J. Booher, W. Liu, A. Petiushko, and A. Garg. "Multi-Constraint Safe RL with Objective Suppression for Safety-Critical Applications". In: <i>Symposium Machine Learning for Autonomous Driving (ML4AD)</i> . 2023
2022	M. Pautov, O. Kuznetsova, N. Tursynbek, A. Petiushko, and I. Oseledets. "Smoothed Embeddings for Certified Few-Shot Learning". In: <i>Advances in Neural Information Processing Systems (NeurIPS)</i> . vol. 35. 2022, pp. 24367–24379
2022	N. Muravev and A. Petiushko. "Certified Robustness via Randomized Smoothing over Multiplicative Parameters of Input Transformations". In: <i>Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, (IJCAI)</i> . 2022, pp. 3366–3372
2021	M. Dzabraev, M. Kalashnikov, S. Komkov, and A. Petiushko. "MDMMT: Multidomain Multimodal Transformer for Video Retrieval". In: <i>Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops</i> . 2021, pp. 3354–3363
2021	S. Komkov and A. Petiushko. "AdvHat: Real-World Adversarial Attack on ArcFace Face ID System". In: <i>2020 25th International Conference on Pattern Recognition (ICPR)</i> . 2021, pp. 819–826
2016	A. Petiushko, D. Parfenov, I. Mazurenko, and A. Kholodenko. <i>Methods and apparatus for merging depth images generated using distinct depth imaging techniques</i> . US Patent App. 14/233,943. 2016
2015	A. Petiushko. "On context-free bigram languages". In: <i>Intelligent Systems</i> 19.2 (2015), pp. 187–208
2010	A. Petiushko. "On Markov Random Fields and Their Relationship with Markov Chains". In: <i>Intelligent Systems</i> 14.1–4 (2010), pp. 225–236

Public Tech Talks

Selected technical talks. For the full list (35+ talks in total), please refer to the [Personal Webpage](#).

Jun 2025	<i>Uncertainty and Safety In Autonomous Driving</i> Invited speaker at 2025 CVPR UCQV Workshop
Mar 2025	<i>ML4AD Award Ceremony</i> Invited speaker and sponsor at 2025 ML4AD Workshop
Jun 2024	<i>Combining Imitation and Reinforcement Learning for Safe Autonomous Driving</i> Invited speaker at 2024 DDADS Workshop
Dec 2023	<i>Scaling Laws for Autonomous Driving Models</i> Invited speaker at 2023 ML4AD Symposium
Jun 2023	<i>Behavior Modeling and Learned Motion Selection for Safe Driving</i> Invited speaker at 2023 CVPR SSAD Workshop
Oct 2022	<i>Autonomy Challenges</i> A Berkeley Deep Drive Lecture
Oct 2021	<i>Certified Robustness, High Dimensions and Computer Vision</i> Invited speaker at SAMPLE - Statistics, Artificial Intelligence, Machine Learning, Probability, Learning Theory Event

Projects

Some projects that I'm proud of.

Jul 2023 - Nov 2023	Nuro ML University (NMLU) The main ideologist and the program owner of Nuro ML University : 100+ full-time employees involved, 3 parallel tracks (novice, average, and SotA).	Nuro, USA/CA, Mountain View
Apr 2023 - ...	Nuro Tech Talks The creator of Nuro Tech Talks series where top robotics researchers share their research ideas for Nuro: 20+ talks with speakers from multiple countries, universities, and companies.	Nuro, USA/CA, Mountain View
Sep 2019 - Dec 2021	School of Huawei Advanced Research Education (SHARE) The main technical coordinator and the owner of Machine Learning and Computer Vision specialization of the School of Huawei Advanced Research Education (briefly SHARE) at Lomonosov Moscow State University, the biggest Russian University.	Huawei Research, Russia, Moscow
Jul 2019	Hackathon: Metric Learning for facial descriptors I was invited by the Russian Association for Artificial Intelligence to organize and run a hackathon to find the best similarity metric for the FaceID system.	Huawei Research, Russia, Moscow
Jan 2017	Python's imresize() I wrote a Python code mimicking the MatLab imresize() function which is often used for super-resolution challenges. Project has apprx 150 stars on github.	Russia, Moscow
Jan 2008	FTP Search Engine I wrote two versions of the ftpsearch engine: one running on the ftp server itself, another for scanning ftp servers remotely; both include an indexer and a web-interface for the search itself. These two versions were deployed on two main Briansk Internet providers of that time: BryanskTel and BKS-TV.	Russia, Briansk

Professional Awards

Moscow Institute of Physics and Technology, Russia, Moscow

2022	Top-10 MIPT Publishing Scientists
Huawei Research, Russia, Moscow	
2020	Best Mentor, Fundamental Research Incentive, Team Golden Medal, Future Star
2019	Outstanding Individual, Future Star, Team Golden Medal, Top Selling Point Delivery
2018	Tech Innovation and Breakthrough, Quality Star, Golden Luban, Excellent Delivery, Customer Success
2017	Outstanding Contractor

Hobbies

Traveling, biking, modern physics, math puzzles, adventure games, family.